Solutions to 2012 Power Question

1. a. Have the Cop stay at A for 2 days. If the Robber is not at A the first day, he must be at one of B_1–B_6, and because the Robber must move along an edge every night, he will be forced to go to A on day 2.

   b. The Cops should stay at $\{A_1, A_3, A_5\}$ for 2 days. If the Robber evades capture the first day, he must have been at an even-numbered hideout. Because he must move, he will be at an odd-numbered hideout the second day. Equivalently, the Cops could stay at $\{A_2, A_4, A_6\}$ for 2 days.

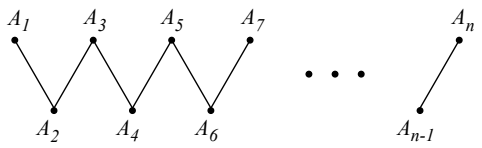
   c. Let $n = h(M)$. The following strategy will always catch a Robber within two days using $n-1$ Cops, which proves that $C(M) \leq n-1$. Choose any subset $\mathcal{S}$ of $n-1$ hideouts and position $n-1$ Cops at the hideouts of $\mathcal{S}$ for 2 days. If the Robber is not caught on the first day, he must have been at the hideout not in $\mathcal{S}$, and therefore must move to a hideout in $\mathcal{S}$ on the following day.

2. a. $C(M) = 6$. The result of 1c applies, so $C(M) \leq 6$. To see that 6 is minimal, note that it is possible for the Robber to go from any hideout to any other hideout in a single night. Suppose that on a given day, the Robber successfully evades capture. Without loss of generality, assume that the Robber was at hideout A_1. Then the only valid conclusion the Cops can draw is that on the next day, the Robber will *not* be at A_1. Thus the only hideout the Cops can afford to leave open is A_1 itself. If they leave another hideout open, the Robber might have chosen to hide there, and the process will repeat.

   b. The following strategy guarantees capture using three Cops for four consecutive days, so $C(M) \leq 3$. Position three Cops at $\{B, E, H\}$ for two days, which will catch any Robber who starts out at B, C, D, E, F, G, or H, because a Robber at C or D would have to move to either B or E, and a Robber at F or G would have to move to either E or H. So if the Robber is not yet caught, he must have been at I, J, K, L, or A for those two days. In this case, an analogous argument shows that placing Cops at B, H, K for two consecutive days will guarantee a capture.

   c. The result in 1c applies, so $C(\mathcal{K}_n) \leq n-1$. It remains to show that $n-1$ is minimal. As was argued in 2a, the Robber always has $n-1$ choices of where to move the next day, so a strategy using less than $n-1$ Cops can never guarantee capture. Thus $C(\mathcal{K}_n) > n-2$.

3. The following argument shows that $C(\mathcal{P}_n) = 1$, and that capture occurs in at most $2n-4$ days. It helps to draw the hideout map as in the following diagram, so that odd-numbered hideouts are all on one level and even-numbered hideouts are all on another level; the case where n is odd is shown below. (A similar argument applies where n is even.)



Each night, the Robber has the choice of moving diagonally left or diagonally right on this map, but he is required to move from top to bottom or vice-versa.

For the first $n-2$ days, the Cop should search hideouts A_2, A_3, ..., A_{n-1}, in that order. For the next $n-2$ days, the Cop should search hideouts A_{n-1}, A_{n-2}, ..., A_2, in that order.

The Cop always moves from top to bottom, or vice-versa, except that he searches A_{n-1} two days in a row. If the Cop is lucky, the Robber chose an even-numbered hideout on the first day. In this case, the Cop and the Robber are always on the same level (top or bottom) for the first half ($n-2$ days) of the search. The Cop is searching from left to right, and the Robber started out to the right of the Cop (or at A_2, the first hideout searched), so eventually the Robber is caught (at A_{n-1}, if not earlier).

If the Robber chose an odd-numbered hideout on the first day, then the Cop and Robber will be on different levels for the first half of the search, but on the same level for the second half. For the second half of the search, the Robber is to the left of the Cop (or possibly at A_{n-1}, which would happen if the Robber moved unwisely or if he spent day $n-2$ at hideout A_n). But now the Cop is searching from right to left, so again, the Cop will eventually catch the Robber.

The same strategy can be justified without using the zig-zag diagram above. Suppose that the Robber is at hideout A_R on a given day, and the Cop searches hideout A_C. Whenever the Cop moves to the next hideout or the preceding hideout, C changes by ± 1, and the Robber's constraint forces R to change by ± 1. Thus if the Cop uses the strategy above, on each of the first $n-2$ days, either the difference $R-C$ stays the same or it decreases by 2. If on the first day $R-C$ is even, either $R-C$ is 0 (the Robber was at A_2 and is caught immediately) or is positive (because $C=2$ and $R \geq 4$). Because the difference $R-C$ is even and decreases (if at all) by 2 each day, it cannot go from positive to negative without being zero. If on the first day $R-C$ is odd, then the Robber avoids capture through day $n-2$ (because $R-C$ is still odd), but then on day $n-1$, R changes (by ± 1) while C does not. So $R-C$ is now even (and is either 0 or negative), and henceforth either remains the same or increases by 2 each day, so again, $R-C$ must be zero at some point between day $n-1$ and day $2n-4$, inclusive.

4. a. $C(\mathcal{C}_n) = 2$. Position one Cop at A_1 and have the Cop search that hideout each day. This strategy reduces $\mathcal{C}_n$ to $\mathcal{P}_{n-1}$, because the Robber cannot get past A_1. Then the second Cop can pursue the strategy from problem 3 on hideouts $\{A_2, \ldots, A_n\}$ to guarantee capture. Thus $C(\mathcal{C}_n) \leq 2$. On the other hand, one Cop is obviously insufficient because the Robber has two choices of where to go every day, so it is impossible for one Cop to cover both possible choices.

   b. If the second Cop makes it all the way to hideout A_n without capturing the Robber, then the second part of the strategy from part (a) can be improved by having the Cops walk back towards each other in a "pincers" move. The details are as follows. If n is odd, then when the second Cop reaches A_n, his and the first Cop's hideouts are both odd. By hypothesis, the Robber's hideout has the opposite (even) parity. (Otherwise the second Cop would have caught the Robber the first time around.) So the two Cops should

each "wait" (that is, search A_1 and A_n again) for one day, after which they will have the same parity as the Robber. Then the first Cop should search hideouts in increasing order while the second Cop searches in decreasing order. This part of the search takes at most $\frac{n-1}{2}$ days, for a total of $(n-1)+1+\frac{n-1}{2}$ days, or $\frac{3n-1}{2}$ days. If n is even, then when the second Cop reaches hideout A_n, the Robber is on an odd hideout (but not A_1), and so the first Cop's hideout is *already* the same parity as the Robber's hideout. So on the next day, the first Cop should search A_2 while the second Cop should re-search A_n. Thereafter, the first Cop searches consecutively increasing hideouts while the second Cop searches consecutively decreasing ones as in the case of n odd. In this case, the search takes $(n-1)+1+\frac{n-2}{2}=\frac{3n-2}{2}$ days.

5. For the map M from 2a, $W(M)=7$. The most efficient strategy is to use 7 Cops to blanket all the hideouts on the first day. Any strategy using fewer than 7 Cops would require 6 Cops on each of two consecutive days: given that any hideout can be reached from any other hideout, leaving more than one hideout unsearched on one day makes it is impossible to eliminate any hideouts the following day. So any other strategy would require a minimum of 12 Cop workdays.

   For the map M from 2b, $W(M)=8$. The strategy outlined in 2b used three Cops for a maximum of four workdays, yielding 12 Cop workdays. The most efficient strategy is to use 4 Cops, positioned at $\{B,E,H,K\}$ for 2 days each. The following argument demonstrates that 8 Cop workdays is in fact minimal. First, notice that there is no advantage to searching one of the hideouts between vertices of the square (for example, C) without searching the other hideout between the same vertices (for example, D). The Robber can reach C on day n if and only if he is at either B or E on day $n-1$, and in either case he could just as well go to D instead of C. So there is no situation in which the Robber is certain to be caught at C rather than at D. Additionally, the Robber's possible locations on day $n+1$ are the same whether he is at C or D on day n, so searching one rather than the other fails to rule out any locations for future days. So any successful strategy that involves searching C should also involve searching D on the same day, and similarly for F and G, I and J, and L and A. On the other hand, if the Robber must be at one of C and D on day n, then he must be at either B or E on day $n+1$, because those are the only adjacent hideouts. So any strategy that involves searching both hideouts of one of the off-the-square pairs on day n is equivalent to a strategy that searches the adjacent on-the-square hideouts on day $n+1$; the two strategies use the same number of Cops for the same number of workdays. Thus the optimal number of Cop workdays can be achieved using strategies that only search the "corner" hideouts B,E,H,K.

   Restricting the search to only those strategies searching corner hideouts B,E,H,K, a total of 8 workdays can be achieved by searching all four hideouts on two consecutive days: if the Robber is at one of the other eight hideouts the first day, he must move to one of the two adjacent corner hideouts the second day. But each of these corner hideouts is adjacent to two other corner hideouts. So if only one hideout is searched, for no matter how many consecutive days, the following day, the Robber could either be back at the previously-searched hideout or be at any other hideout: no possibilities are ruled out. If two adjacent corners are searched, the Cops do no better, as the following argument shows. Suppose that B and E are both

searched for two consecutive days. Then the Cops can rule out B, E, C, and D as possible locations, but if the Cops then switch to searching either H or K instead of B or E, the Robber can go back to C or D within two days. So searching two adjacent corner hideouts for two days is fruitless and costs four Cop workdays. Searching diagonally opposite corner hideouts is even less fruitful, because doing so rules out none of the other hideouts as possible Robber locations. Using three Cops each day, it is easy to imagine scenarios in which the Robber evades capture for three days before being caught: for example, if B, E, H are searched for two consecutive days, the Robber goes from I or J to K to L. Therefore if three Cops are used, four days are required for a total of 12 Cop workdays.

If there are more than four Cops, the preceding arguments show that the number of Cops must be even to produce optimal results (because there is no advantage to searching one hideout between vertices of the square without searching the other). Using six Cops with four at corner hideouts yields no improvement, because the following day, the Robber could get to any of the four corner hideouts, requiring at least four Cops the second day, for ten Cop workdays. If two or fewer Cops are at corner hideouts, the situation is even worse, because if the Robber is not caught that day, he has at least nine possible hideouts the following day (depending on whether the unsearched corners are adjacent or diagonally opposite to each other). Using eight Cops (with four at corner vertices) could eliminate one corner vertex as a possible location for the second day (if the non-corner hideouts searched are on adjacent sides of the square), but eight Cop workdays have already been used on the first day. So 8 Cop workdays is minimal.

6. A single Cop can only search one hideout in a day, so as long as M has two or more hideouts, there is no strategy that guarantees that a lone Cop captures the Robber the first day. Then either more than one Cop will have to search on the first day, or a lone Cop will have to search for at least 2 days; in either of these cases, $W(M) \geq 2$. On the other hand, n Cops can always guarantee a capture simply by searching all n hideouts on the first day, so $W(M) \leq n$.

   To show that the lower bound cannot be improved, consider the *star on n hideouts*; that is, the map $\mathcal{S}_n$ with one central hideout connected to $n-1$ outer hideouts, none of which is connected to any other hideout. (The map in 1a is $\mathcal{S}_7$.) Then $W(\mathcal{S}_n) = 2$, because a single Cop can search the central hideout for 2 days; if the Robber is at one of the outer hideouts on the first day, he must go to the central hideout on the second day. For the upper bound, the complete map $\mathcal{K}_n$ is an example of a map with $W(M) = n$. As argued above, the minimum number of Cops needed to guarantee a catch is $n-1$, and using $n-1$ Cops requires two days, for a total of $2n-2$ Cop workdays. So the optimal strategy for $\mathcal{K}_n$ (see 2c) is to use n Cops for a single day.

7. a. Suppose M is bipartite, and let $\mathcal{A}$ and $\mathcal{B}$ be the sets of hideouts referenced in the definition. Because $\mathcal{A}$ and $\mathcal{B}$ are disjoint, either $|\mathcal{A}| \leq n/2$ or $|\mathcal{B}| \leq n/2$ or both. Without loss of generality, suppose that $|\mathcal{A}| \leq n/2$. Then position Cops at each hideout in $\mathcal{A}$ for two days. If the Robber was initially on a hideout in $\mathcal{B}$, he must move the following day, and because no hideout in $\mathcal{B}$ is connected to any other hideout in $\mathcal{B}$, his new hideout must be a hideout in $\mathcal{A}$.

b. The given condition actually implies that the graph is bipartite. Let A_1 be a hideout in M, and let $\mathcal{A}$ be the set of all hideouts V such that all paths from A_1 to V have an even number of edges, as well as A_1 itself; let $\mathcal{B}$ be the set of all other hideouts in M. Notice that there are no edges from A_1 to any other element A_i in $\mathcal{A}$, because if there were, that edge would create a path from A_1 to A_i with an odd number of edges (namely 1). So A_1 has edges only to hideouts in $\mathcal{B}$. In fact, there can be no edge from any hideout A_i in $\mathcal{A}$ to any other hideout A_j in $\mathcal{A}$, because if there were, there would be a path with an odd number of edges from A_1 to A_j via A_i. Similarly, there can be no edge from any hideout B_i in $\mathcal{B}$ to any other hideout B_j in $\mathcal{B}$, because if there were such an edge, there would be a path from A_1 to B_j with an even number of edges via B_i. Thus hideouts in $\mathcal{B}$ are connected only to hideouts in $\mathcal{A}$ and vice versa; hence M is bipartite.

8. Because a Robber in $\mathcal{A}_i$ can only move to a hideout in $\mathcal{A}_{i-1}$ or $\mathcal{A}_{i+1}$, this map is essentially the same as the cyclic map $\mathcal{C}_k$. So the Cops should apply a similar strategy. First, position n/k Cops at the hideouts of set $\mathcal{A}_1$ and n/k Cops at the hideouts of set $\mathcal{A}_2$. On day 2, leave the first n/k Cops at $\mathcal{A}_1$, but move the second group of Cops to $\mathcal{A}_3$. Continue until the second set of Cops is at $\mathcal{A}_k$; then "wait" one turn (search $\mathcal{A}_k$ again), and then search backward $\mathcal{A}_{k-1}$, $\mathcal{A}_{k-2}$, etc.

9. There are many examples. Perhaps the simplest to describe is the *complete bipartite map* on the hideouts $\mathcal{A} = \{A_1, A_2, \ldots, A_{17}\}$ and $\mathcal{B} = \{B_1, B_2, \ldots, B_{1995}\}$, which is often denoted $M = \mathcal{K}_{17,1995}$. That is, let M be the map with hideouts $\mathcal{A} \cup \mathcal{B}$ such that A_i is adjacent to B_j for all i and j, and that A_i is not adjacent to A_j, nor is B_i adjacent to B_j, for any i and j.

   If 17 Cops search the 17 hideouts in $\mathcal{A}$ for two consecutive days, then they are guaranteed to catch the Robber. This shows that $C(M) \leq 17$ and that $W(M) \leq 34$. If fewer than 17 Cops search on a given day, then the Robber has at least one safe hideout: he has exactly one choice if the Cops search 16 of the hideouts A_i and the Robber was hiding at some B_j the previous day, 1995 choices if the Robber was hiding at some A_i the previous day. Thus $C(M) = 17$.

   On the other hand, fewer than 34 Cop workdays cannot guarantee catching the Robber. Unless the Cops search every A_i (or every B_j) on a given day, they gain no information about where the Robber is (unless he is unlucky enough to be caught).

   The complete bipartite map is not the simplest in terms of the number of edges. Try to find a map with 2012 hideouts and as few edges as possible that has a Cop number of 17 and a workday number of 34.

10. No such map exists. Let M be a map with at least four hideouts. The proof below shows that either $W(M) > 3$ or else M is a star (as in the solution to problem 6), in which case $W(M) = 2$.

    First, suppose that there are four distinct hideouts A_1, A_2, B_1, and B_2 such that A_1 and A_2 are adjacent, as are B_1 and B_2. If the Cops make fewer than four searches, then they cannot search $\{A_1, A_2\}$ twice and also search $\{B_1, B_2\}$ twice. Without loss of generality, assume the Cops search $\{A_1, A_2\}$ at most once. Then the Robber can evade capture by moving from A_1 to A_2 and back again, provided that he is lucky enough to start off at the right one. Thus

$W(M) > 3$ in this case. For the second case, assume that whenever A_1 is adjacent to A_2 and B_1 is adjacent to B_2, the four hideouts are *not* distinct. Start with two adjacent hideouts, and call them A_1 and A_2. Consider any hideout, say B, that is not one of these two. Then B must be adjacent to some hideout, say C. By assumption, A_1, A_2, B, and C are not distinct, so $C = A_1$ or $C = A_2$. That is, every hideout B is adjacent to A_1 or to A_2.

If there are hideouts B_1 and B_2, distinct from A_1 and A_2 and from each other, such that B_1 is adjacent to A_1 and B_2 is adjacent to A_2, then it clearly violates the assumption of this case. That is, either every hideout B, distinct from A_1 and A_2, is adjacent to A_1, or every such hideout is adjacent to A_2. Without loss of generality, assume the former.

Because every hideout is adjacent to A_1 (except for A_1 itself), the foregoing proves that the map M is a star. It remains to show that there are no "extra" edges in the map. For the sake of contradiction, suppose that B_1 and B_2 are adjacent hideouts, both distinct from A_1. Because the map has at least four hideouts, choose one distinct from these three and call it A_2. Then these four hideouts violate the assumption of this case.